石荣晔

北京航空航天大学，副教授，博士生导师
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教育背景

Sep. 2008—Jul.2012 **学士,中国农业大学,北京,中国**
专业：电子信息科学与技术

Sep. 2012—Jul.2014 **硕士,北京大学,北京,中国**
专业：电子与通信工程

Sep. 2014—May.2019 **博士,电子与计算机工程系,卡内基梅隆大学(Carnegie Mellon University),美国**
专业：电子与计算机工程
导师：Manuela M. Veloso院士（美国工程院院士，IEEE/ACM/AAAI/AAAS Fellow）
Peter Steenkiste教授（IEEE Fellow）

工作经历

Sep. 2019—Aug.2020 **博士后研究员,工学院,美国哥伦比亚大学(Columbia University),美国**
导师：Sharon (Xuan) Di副教授
Qiang Du教授（AAAS Fellow，美国数学会Fellow）
- 研究基于物理信息神经网络等混合模型进行交通态的估值和车辆跟驰模型的学习

Sep. 2020—Jan. 2023 **主任工程师(天才少年计划),华为技术有限公司,深圳,中国**
- 智能交通信控产品TrafficGo的AI信控算法负责人
- 2012实验室中央研究院交通系统建模与算法领域“助理首席专家”

March. 2023—present **副教授,博导,人工智能学院（人工智能研究院）,北京航空航天大学**
- 研究群体智能、多智能体强化学习、物理信息神经网络、AI for Science等

获奖情况

入选政府人才计划：
- 2024 北京市科技新星计划，北京市科学技术委员会、中关村科技园区管理委员会，省部级人才计划 (1/1)
- 2021 深圳市“孔雀计划”海外高层次 C 类人才，深圳市人力资源和社会保障局 (1/1)
- 2022 深圳市龙岗区“深龙英才”人才计划（C 类），深圳市龙岗区人力资源局 (1/1)

国际奖项：
- 2019 美国亚马逊 AWS 机器学习研究奖 (3/3)
- GLSVLSI 2017 会议最佳论文奖(3/5)
- 美国自然基金委员会(NSF) Travel 奖 (1/1)
- 2022 ICML (CCF A 类会议) 杰出审稿人(前 10%) (1/1)
- 2015 Sonic John Bardeen 研究奖 (1/3)
- 2014 Google 优秀奖学金 (1/1)

企业荣誉：
- 2020 入选华为“天才少年”计划 (1/1)

- 2022 华为中央研究院交通系统建模与算法领域“助理首席专家”称号(任正非签发）(1/1)
- 2022 获华为“创新先锋”总裁奖二等奖 (1/1)

国内学术奖：

- 2022 首届“北京智慧交通开放创新大赛”算法赛道**一等奖**(北京市经信局、交通委员会、公安局公安交通管理局主办) (6/10)
- 2025 首届上海人工智能实验室明珠湖会议最佳提案奖(浦江国家实验室) (1/1)

教学类奖项：

- 2024 第三届“余姚杯”中国高校机器人实验教学创新大赛三等奖 (1/4)
- 北京市普通高校本科毕设优秀指导教师，北京市教育委员会 (1/1)

科研项目情况

- 科技创新 2030-“新一代人工智能”重大项目，**国家级纵向**，人工智能科学计算范式与共性平台体系，2023.3-2026.2，**子课题负责人（经费 120 万）**
- 国家自然科学基金青年科学基金项目，**国家级纵向**，基于机理驱动源项估计和路径规划的多智能体强化学习寻源方法研究，2024.1-2026.12，**项目负责人（经费 30 万）**
- 北京市科技新星计划项目，**省部级纵向**，复杂异构群体智能度量理论与调控方法，2024.9-2027.9，**项目负责人（经费 40 万）**
- 科技创新 2030-“新一代人工智能”重大项目，**国家级纵向**，数理知识驱动的小行星在线智能计算框架，2023.3-2027.2，参与 **（经费 50 万）**
- 北京航空航天大学“敢为计划”项目，校级项目，内嵌数理的群体具身智能方法研究，2024.1-2025.12，**项目负责人（经费 20 万）**
- 航天三院 304 所技术服务项目，横向项目，感知模型训练数据采集，2023.2~2023.12，**项目负责人（经费 24 万）**已结项
- 美国亚马逊 AWS 机器学习研究项目，横向项目，Multi-Autonomous Vehicle Driving Policy Learning for Efficient and Safe Traffic，2020.6-2021.6，**Co-PI（经费 90.3 万）**已结项

科研项目情况

- 教育部-腾讯产学合作协同育人项目，省部级教改项目，基于强化学习的智能决策人工智能课程建设，2024.10-2025.10，**项目负责人（经费 5 万）**

个人简介

石荣晔现任北京航空航天大学人工智能学院副教授，博士生导师，2019 年从美国卡耐基梅隆大学电子与计算机工程专业获得博士学位，师从 Manuela M. Veloso (美国工程院院士，IEEE/ACM/AAAI/AAAS Fellow)和 Peter Steenkiste (IEEE Fellow)。2019~2020 在美国哥伦比亚大学担任 Postdoctoral Research Scientist，师从 Xuan Di 和 Qiang Du (AAAS Fellow)。回国后曾任职于华为，入选北京市科技新星、华为“天才少年”计划和深圳市“孔雀计划”海外高层次人才。

长期从事领域知识内嵌人工智能算法方面的研究，在物理信息神经网络、多智能体系统、强化学习及其在智慧城市、AI for Science 领域的应用等方面开展了许多工作，以“领域知识+人工智能”为创新导向，从数据效率、可解释性、可拓展性等维度解决人工智能在智慧城市场景中融合深度不足、应用广度受限的问题。已发表 SCI/EI 学术论文 40 余篇，多篇以第一作者或通讯作者身份发表在 AAAI、IEEE TITS、Information Processing & Management、IEEE TASE 等人工智能领域的顶级会议/期刊上，曾获 Amazon AWS Machine Learning Research Award、GLSVLSI 2017 会议最佳论文奖、NSF Travel Award、John Bardeen Research Award、ICML (Top 10%) Outstanding Reviewers 等奖项，相关工作被福布斯(Forbes)、ScienceDaily、机器之心等主

流科技资讯平台报道，曾被华为任命为交通系统建模与算法领域的“助理首席专家”并获授华为“创新先锋”总裁奖。主持科技部-科技创新 2030-“新一代人工智能”重大项目子课题、国自然青基等项目。

曾担任 ECML-PKDD 2021、IEEE ICHMS 2020/2021 等会议程序委员会委员，ITSC 2018 Regular Session 共同主席，担任 Proceedings of the IEEE、IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)、IEEE Transactions on Neural Networks and Learning Systems (TNNLS)等国际顶级期刊审稿人，以及 ICML、NeurIPS、KDD 等国际会议审稿人。担任 IEEE 和 ACM 会员，CCF、中国图象图形学会会员及专委会委员等。

教学方面担任本科生、研究生毕设指导教师，曾获评北京市普通高校本科毕设优秀指导教师；承担群体智能等核心专业课，获第三届“余姚杯”中国高校机器人实验教学创新大赛三等奖，主持产学合作协同育人项目。

学术评价与应用成就

本人研究成果获得了国内外院士、IEEE/ACM Fellow 等国际著名专家学者的引用与正面评价。例如，美国卡耐基梅隆大学的 José M. F. Moura 教授（美国工程院院士、IEEE Life Fellow、前 IEEE 主席）在其学术评论函件中指出：本人的研究**显著缩小了 MARL 算法与实际应用之间的差距**（Rongye’s research significantly narrows the gap between MARL and its practical application to real world problems）。美国摩根大通集团人工智能研究主任 Manuela M. Veloso 教授（美国工程院院士、IEEE/ACM/AAAI/AAAS Fellow、前 AAAI 主席）在其学术评论函件中认为本人提出的 SC-M*算法在解决复杂优化问题方面**是一个成功且引人注目的成果**（the SC-M* algorithm is remarkably successful in combining and optimizing for the multiple complexities of the problem）。香港城市大学 Min Xie 教授（欧洲科学与艺术院院士，IEEE Fellow）等人在其论文（T-ITS，2023）中指出：本人的工作**被广泛应用于解决数据稀疏的问题**（to address above problems (data is rare), the hybrid methods combining physics-based and data driven-based methods have been widely applied）。此外，本人积极服务政府相关部门，部分成果获应用成效，在工业界中获得了应用。例如所提出的 OBLLISC 算法赋能华为 TrafficGo 多路口信控产品，在深圳侨香路主干道下发实测，效果显著，获得深圳交管部门的“**AI 信控高峰期明显，平峰有**

效果”、“**成果显著**”等好评，产生良好社会和经济效益。面向交通环境态势认知场景，提出基于路口关联先验特性的流量预测和补全算法，获北京市经信局、交通委员会、公安局交管局主办的**首届北京智慧交通开放创新大赛一等奖**，成果将优先应用于北京市智慧交通领域相关应用。所提出的“一种 PINN 融合传统数值计算高效求解大规模微分方程的方法”曾助力华为煤矿军团，解决煤矿行业物探的地质反演痛点问题，获**华为“创新先锋”总裁奖二等奖**，产生良好商业价值。本人曾受国家消防救援局邀请，进行以“人工智能发展概况及未来应用前景展望”为主题的咨询报告和专题授课，被该局给予高度评价，认为**“内容丰富、信息量大…做了非常专业的解读”，获得国家消防救援局书面感谢信 1 封**，并已开启群体智能赋能消防救援、灾害应急响应等技术合作洽谈。

部分学术服务情况

- IEEE会员
- ACM会员
- 中国计算机学会会员
- 中国图象图形学会会员
- 中国图像图形学会空间信息感知与决策专委会委员
- 担任ITSC 2018国际会议联合分会主席
- SCI期刊Electronics特刊客座编辑
- 部分顶级期刊审稿情况
  - Proceedings of the IEEE
  - ACM Computing Surveys
  - IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
  - IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
  - IEEE Transactions on Intelligent Transportation Systems (TITS)
  - IEEE Transactions on Mobile Computing (TMC)
- 部分顶级会议审稿情况
  - The 38th International Conference on Machine Learning (ICML 2021)
  - The 34th Annual Conference on Neural Information Processing Systems (NeurIPS 2020)
  - ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2021)
  - European Conference on Machine Learning (ECML-PKDD 2021)
  - IEEE International Conference on Human-Machine Systems (ICHMS 2020)
- 担任会议PC/TPC成员
  - ECML-PKDD 2021、2022
  - ICHMS 2020
  - SMART 2019
- 其它
  -2024年度全国研究生教育评估监测专家库专家（教育部学位与研究生教育发展中心）
  -2024年度国家自然科学基金委员会项目评审专家
  -2023年国家消防救援局“人工智能发展概况及未来应用前景展望”专题授课专家

学术论文发表和专利情况

（#为共同第一作者，*为通讯作者，CCF 为中国计算机学会，CAA 为中国自动化学会，CAAI 为中国人工智能学会）：

[1] **Rongye Shi**, Xin Yu, Yandong Wang, Yongkai Tian, Zhenyu Liu, Wenjun Wu, Xiao-Ping Zhang, and Manuela M. Veloso. "Symmetry-Informed MARL: A Decentralized and Cooperative UAV Swarm Control Approach for Communication Coverage." ***IEEE Transactions on Mobile Computing***, 2025, pp. 1-18. **(中科院 SCI 一区 Top、CCF-A 类期刊、IF=9.2)**

[2] **Rongye Shi**, Zhaobin Mo, and Xuan Di. "Physics-Informed Deep Learning for Traffic State Estimation: A Hybrid Paradigm Informed by Second-Order Traffic Models." ***AAAI Conference on Artificial Intelligence (AAAI)***, 2021, pp. 540-547. **(CCF-A 类、CAAI-A 类会议)**

[3] **Rongye Shi**, Zhaobin Mo, Kuang Huang, Xuan Di*, and Qiang Du. "A Physics-Informed Deep Learning Paradigm for Traffic State and Fundamental Diagram Estimation." ***IEEE Transactions on Intelligent Transportation Systems***, vol. 23, no. 8, 2022, pp. 11688-11698. **(中科院 SCI 一区 Top、CAA-A 类、CCF-B 类期刊、IF=8.4)**

[4] **Rongye Shi***, Peter Steenkiste, and Manuela M. Veloso. "Improving the On-Vehicle Experience of Passengers Through SC-M*: A Scalable Multi-Passenger Multi-Criteria Mobility Planner." ***IEEE Transactions on Intelligent Transportation Systems***, vol. 22, no. 2, 2021, pp. 1026-1040. **(中科院 SCI 一区 Top、CAA-A 类、CCF-B 类期刊、IF=8.4)**

[5] **Rongye Shi**#, Ji Luo#, Nan Zhou, Yuxuan Liu, Chaopeng Hong, Xiao-Ping Zhang, and Xinlei Chen*. "Phy-APMR: A Physics-Informed Air Pollution Map Reconstruction Approach with Mobile Crwod-Sensing for Fine-Grained Measurement." ***Building and Environment***, vol. 272, 2025, pp. 1-15. **(中科院 SCI 一区 Top 期刊、IF=7.6)**

[6] **Rongye Shi**, Peter Steenkiste, and Manuela Veloso. "Generating Synthetic Passenger Data through Joint Traffic-Passenger Modeling and Simulation." ***International Conference on Intelligent Transportation Systems (ITSC)***, 2018, pp. 3397-3402. **(CAA-A 类会议)**

[7] Xiaolin Liu#, **Rongye Shi**#, Qianxin Hui, Susu Xu, Shuai Wang, Rui Na, Ying Sun, Wenbo Ding, Dezhi Zheng*, and Xinlei Chen*. "TCACNet: Temporal and Channel Attention Convolutional Network for Motor Imagery Classification of EEG-based BCI." ***Information Processing and Management***, vol. 59, no. 5, 2022, pp.1-17. **(中科院 SCI 一区 Top、CCF-B 类期刊、IF=6.9)**

[8] Pu Feng, **Rongye Shi***, Size Wang, Qizhen Wu, Xin Yu, and Wenjun Wu. "Lyapunov-Informed Multi-Agent Reinforcement Learning for Multi-Robot Cooperation Tasks." ***IEEE Transactions on Automation Science and Engineering***, vol. 22, 2025, pp. 20263-20279. **(CAA-A+类、CCF-B 类期刊，中科院 SCI 二区 Top 期刊、IF=6.4)**

[9] Yu, Xin, **Rongye Shi***, Pu Feng, Yongkai Tian, Simin Li, Shuhao Liao, and Wenjun Wu. "Leveraging Partial Symmetry for Multi-Agent Reinforcement Learning." ***AAAI Conference on Artificial Intelligence (AAAI)***, 2024, pp. 17583-17590. **(CCF-A 类、CAAI-A 类会议)**

[10] Chen Hou, **Rongye Shi***, Qilong Huang*, and Yifang Wang. "Energy Harvest of Multiple Smart Sensors with Real-Time Fault-Detection." ***IEEE Transactions on Automation Science and Engineering***, vol. 21, no. 2, 2024, pp. 1592-1606. **(CAA-A+类、CCF-B 类期刊，中科院 SCI 二区 Top 期刊、IF=6.4)**

[11] Yanjun Pu, Fang Liu, **Rongye Shi***, Haitao Yuan*, Ruibo Chen, Tianhao Peng, and Wenjun Wu*. "ELAKT: Enhancing Locality for Attentive Knowledge Tracing." ***ACM Transactions on Information Systems***, vol. 42, no. 4, 2024, pp. 1-27. **(CCF-A 类、CAAI-A 类期刊，中科院 SCI 二区期刊、IF=9.1)**

[12] Xin Yu, **Rongye Shi***, Pu Feng, Yongkai Tian, Jie Luo, and Wenjun Wu. "ESP: Exploiting Symmetry Prior for Multi-Agent Reinforcement Learning." ***European Conference on Artificial Intelligence (ECAI)***, 2023, pp. 2946-2953. **(CCF-B 类会议)**

[13] Cristian Axenie#, **Rongye Shi**#,*, Daniele Foroni#, Alexander Wieder#, Mohamad Al Hajj Hassan#, Paolo Sottovia#, Margherita Grossi#, Stefano Bortoli#, and Götz Brasche. "OBELISC: Oscillator-Based Modelling and Control Using Efficient Neural Learning for Intelligent Road Traffic Signal Calculation." ***European Conference on Machine Learning (ECML-PKDD)***, 2021, pp. 437-452. **(CCF-B 类会议)**

[14] Xin Yu, Yongkai Tian, Li Wang, Pu Feng, Wenjun Wu, and **Rongye Shi***. "AdaptAUG: Adaptive Data Augmentation Framework for Multi-Agent Reinforcement Learning." ***IEEE International Conference on Robotics and Automation (ICRA)***, 2024, pp. 10814-10820. **(CAA-A 类、CCF-B 类会议)**

[15] Jiabin Lou, Wenjun Wu, Shuhao Liao, and **Rongye Shi***. "Air-M: A Visual Reality Many-Agent Reinforcement Learning Platform for Large-Scale Aerial Unmanned System." ***IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)***, 2023, pp. 5598-5605. **(CAA-A 类会议)**

[16] Shuhao Liao, Xuehong Liu, Wenjun Wu, **Rongye Shi***, Junyu Zhang, and Haopeng Wang. "Synaptic Weight Optimization for Oscillatory Neural Networks: A Multi-Agent RL Approach." ***IEEE International Conference on Agents (ICA)***, 2024, pp. 152-158. **(EI)**

[17] **Rongye Shi***, Peter Steenkiste*, and Manuela M. Veloso*. "SC-M*: A Multi-Agent Path Planning Algorithm with Soft-Collision Constraint on Allocation of Common Resources." ***Applied Sciences (Appl. Sci.)***, 2019. (SCI)

[18] Zhicheng Li, Zhihao Gu, Xuan Di, and **Rongye Shi***. "An LSTM-Based Autonomous Driving Model Using a Waymo Open Dataset." ***Applied Sciences (Appl. Sci.)***, 2020. (SCI)

[19] **Rongye Shi**, Peter Steenkiste, and Manuela Veloso. "Second-Order Destination Inference using Semi-Supervised Self-Training for Entry-Only Passenger Data." ***IEEE/ACM 10th International Conference on Big Data Computing, Applications and Technologies (BDCAT)***, 2017. **(EI、获 NSF Travel 奖)**

[20] **Rongye Shi**, Thomas Jackson, Brian Swenson, Soummya Kar, and Lawrence Pileggi. "On the Design of Phase Locked Loop Oscillatory Neural Networks: Mitigation of Transmission Delay Effects." ***International Joint Conference on Neural Networks (IJCNN)***, 2016. **(EI、CCF-C 类会议)**

[21] **Shi Rong-Ye**, and Wang Yan-Hui*. "Analysis of Influence of RF Power and Buffer Gas Pressure on Sensitivity of Optically Pumped Cesium Magnetometer." ***Chinese Physics B (Chin. Phys. B)***, 2013. (SCI)

[22] Cristian Axenie#, **Rongye Shi#,***, Daniele Foroni#, Alexander Wieder#, Mohamad Al Hajj Hassan#, Paolo Sottovia#, Margherita Grossi#, Stefano Bortoli#, and Götz Brasche. "TRAMESINO: Traffic Memory System for Intelligent Optimization of Road Traffic Control." ***Advanced Analytics and Learning on Temporal Data: 6th ECML PKDD Workshop (AALTD)***, 2021. (EI)

[23] **Rongye Shi**, Chang Liu, Sheng Zhou, and Yanhui Wang. "Digital-Locking Optically Pumped Cesium Magnetometer." ***The European Frequency and Time Forum (EFTF)***, 2014. (EI)

[24] 宋琪, 陈天宇, 王子铭, 金圣凯, 高崇涵, **石荣晔***, 周号益, 李建欣*. "基于情境化神经算子的空气动力学风阻预测." ***人工智能***, vol. 5, 2024, pp. 1-10. **(中国核心期刊)**

[25] Pu Feng, **Rongye Shi**, Size Wang, Junkang Liang, Xin Yu, Simin Li, and Wenjun Wu*. "Safe and Efficient Multi-Agent Collision Avoidance with Physics-Informed Reinforcement Learning." ***IEEE Robotics and Automation Letters***, vol. 9, no. 12, 2024, pp. 11138-11145. **(CAAI-B 类期刊，中科院 SCI 二区 Top 期刊、IF=5.3)**

[26] Xuan Di*, and **Rongye Shi**. "A Survey on Autonomous Vehicle Control in the Era of Mixed-Autonomy: From Physics-Based to AI-Guided Driving Policy Learning." ***Transportation Research Part C: Emerging Technologies***, 2021. **(中科院 SCI 一区 Top 期刊，CAA-A 类期刊、IF=7.9)**

[27] Baining Zhao#, Xuzhe Wang#, Tianyu Zhang, **Rongye Shi**, Fengli Xu, Fanhang Man, Erbing Chen, Yang Li, Yong Li, Tao Sun, and Xinlei Chen*. "Estimating and Modeling Spontaneous Mobility Changes during the COVID-19 Pandemic Without Stay-at-Home Orders." ***Humanities and Social Sciences Communications (HSSC)***, 2024. **(Nature 子刊)**

[28] Zhaobin Mo, **Rongye Shi**, and Xuan Di*."A Physics-Informed Deep Learning Paradigm for Car-Following Models." ***Transportation Research Part C: Emerging Technologies***, 2021. **(中科院 SCI 一区 Top 期刊，CAA-A 类期刊、IF=7.9)**

[29] Jiabin Lou, **Rongye Shi**, Yuxin Lin, Qunbo Wang, and Wenjun Wu*. "TALKER: A Task-Activated Language Model Based Knowledge-Extension Reasoning System." ***IEEE Robotics and Automation Letters***, vol. 10, no. 2, 2025, pp. 1026-1033. **(CAAI-B 类期刊，中科院 SCI 二区 Top 期刊、IF=5.3)**

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